

Aayush Arya

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SUMMARY

Computer Science undergraduate (B.Tech, AI specialization) currently working as a Software Engineering Intern at Mycom, building production backend services in Spring Boot / Java for a reactive microservices platform. Combines hands-on backend development, cloud infrastructure (AWS), and strong CS fundamentals to deliver reliable, well-tested software as a Software Development Engineer.

EDUCATION

Dronacharya College of Engineering, Gurugram

Aug 2023 - May 2027

B.Tech in Computer Science & Engineering (Artificial Intelligence) CGPA: 7.1 / 10 (through Semester IV)

12th (CBSE): 72.33%

10th (CBSE): 86.8%

Relevant Coursework: Design & Analysis of Algorithms, Data Structures and Algorithms, Operating Systems, Computer Networks, Computer Architecture, Relational Databases, Artificial Intelligence, Machine Learning & Pattern Recognition, Graph Theory, Probability & Statistics

TECHNICAL SKILLS

Core CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems (DBMS), Graph Theory & Complexity Theory

Languages: Java, Python, C, SQL, R

Backend, APIs & Data: Spring Boot 3, Spring WebFlux/Project Reactor, REST APIs (JAX-RS), Spring Security, OAuth2/OIDC, MongoDB (Reactive), MySQL, Caffeine Cache

Testing & Quality: JUnit 5, Mockito, Cucumber BDD, Maven Surefire, Code Review

Cloud, DevOps & Tools: AWS (EC2, S3, Lambda, EKS, IAM, CloudWatch), GitLab CI, Git, GitHub, Maven, Jira, IntelliJ IDEA, VS Code

Machine Learning & Data: TensorFlow, Keras, PyTorch (beginner), Scikit-learn, Pandas, NumPy, OpenCV, Matplotlib

EXPERIENCE

Software Engineering Intern — Mycom

Jun 2026 - Aug 2026

PrOptima V6.0, Content Manager Service | Onsite

- Own end-to-end backend implementation of the Embedded View feature for PrOptima's Content Manager service (eaa-proptima-web-reports) - a Spring Boot 3 / WebFlux (Java 21) reactive microservice enabling operators to embed external URLs, text, and images directly into product dashboards.
- Designed and implemented a 4-layer data pipeline (DTO -> App Model -> Entity -> PTO -> MongoDB), introducing ReportType.EMBEDDED as a first-class view type across 5 new classes and 15+ modified files, mirroring the existing Gauge/Scorecard pattern with zero added tech debt.
- Engineered production-grade validation and security controls - SSRF protection, image MIME/size restrictions, and single-mode validation rules - backed by a Caffeine cache (30-minute TTL) with mutation-based eviction.
- Delivered 10+ JUnit/Mockito unit tests and 10 passing Cucumber BDD scenarios plus OpenAPI 3 documentation, resolving all code-review findings across 2 completed sprints and unblocking the downstream frontend integration team.
- Completed the project's final two sprints - wiring the Embedded View panel to persist and reload configurations end-to-end, and delivering render-time dynamic URL token resolution with security hardening (URL allow-listing, sandboxed iframe/CSP, image validation) for production rollout.

AWS Cloud Engineering Intern — Maruti Suzuki India Limited

Jun 2025 - Jul 2025

Onsite, India

- Implemented IAM policies and access-control levels for users and groups, enforcing least-privilege security standards across the organization's AWS environment.
- Automated infrastructure provisioning and management using AWS services including EC2, S3, EKS, and Lambda, streamlining deployment of cloud-based workloads.
- Configured CloudWatch Logs, Metrics, and Alarms to monitor resource utilization and system health across multiple cloud deployments.

PROJECTS

InterviewPilot — AI Interview Copilot

Python (FastAPI), TypeScript/React, PostgreSQL, Redis, Claude API, Docker, GitHub Actions

- Built a production-grade SaaS platform for technical interview preparation - AI mock interviews, resume analysis, and a Monaco-editor coding round with real code execution and AI-driven review - powered by the Claude API across 20+ tracks.
- Architected the FastAPI backend (clean architecture, SQLAlchemy ORM) with JWT authentication (Google/GitHub OAuth) and containerized the full stack (React, PostgreSQL, Redis, Nginx) via Docker Compose with GitHub Actions CI/CD.

NextRec — AI-Powered Personalized Recommendation System

Python (FastAPI), React, scikit-learn, SciPy, Docker, GitHub Actions

- Built a full-stack recommendation platform combining content-based filtering (TF-IDF/cosine similarity), collaborative filtering (SVD matrix factorization), and semantic search (sentence-transformers/FAISS) across 6 content domains.
- Designed a hybrid recommendation engine fusing content, collaborative, trending, and popularity signals via weighted scoring, plus a cold-start solver for new users; deployed via Docker Compose and GitHub Actions CI/CD to Render.

SentinelDesk — Enterprise Customer Support & SLA Desk SaaS Platform

NestJS, Prisma, PostgreSQL, Next.js 15, React 19, Tailwind v4, Redis, Docker Compose

- Built a from-scratch monorepo support/ticketing platform with multi-tenant auth, JWT access/refresh rotation, and a real five-role RBAC system (Role/Permission schema + guards), backed by Helmet, rate limiting, CSRF protection, and audit logging.

CloudOptix AI — AI-Powered AWS Cost Optimization Platform

FastAPI, React, TypeScript, PostgreSQL, boto3, scikit-learn, Docker

- Built a self-hosted alternative to AWS Cost Explorer/Trusted Advisor/Compute Optimizer that scans EC2, S3, Lambda, IAM, and CloudTrail via read-only credentials, running a custom rule-based recommendation engine, scikit-learn forecast engine, and cost simulator using only Free Tier-eligible API calls.

Smart Attendance AI — AI-Powered Attendance & Workforce Management Platform

Next.js 15, FastAPI, InsightFace/ONNX Runtime, PostgreSQL + pgvector, Celery, Redis, Docker, Terraform

- Built a self-hosted face-recognition attendance pipeline (RetinaFace + ArcFace embeddings, pgvector HNSW search) with liveness detection, anti-spoof classification, composite fraud scoring, geofencing, RBAC, and a WebSocket-pushed live dashboard.

ACHIEVEMENTS

- Advanced to the semifinal round of Unstop AtomQuest (18,000+ registrants) with a cleaning-robot solution.

CERTIFICATIONS

IIT Bombay - C Programming (Dec 2023) | NPTEL - Java (Apr 2025) | Infosys Springboard - Python & AI (Feb 2025)

IBM - Generative AI for Software Development (Feb 2025) | AWS Cloud Practitioner Essentials (Jul 2025) | OCI 2025 Foundations Associate (Nov 2025)